

Final Project

Nguyen Van Thanh

ID: 20242084

1 Problem 1: Cart-Pole with Actuator Dynamics: DDP and iLQR

Prerequisite: cart-Pole Dynamics

The dynamic equation of cart-pole system is as follows:

$$\ddot{x} = \frac{\ell m_p \sin(\theta) \dot{\theta}^2 + F + m_p g \cos(\theta) \sin(\theta)}{m_c + m_p (1 - \cos^2(\theta))} \quad (1.1)$$

$$\ddot{\theta} = -\frac{\ell m_p \cos(\theta) \sin(\theta) \dot{\theta}^2 + F \cos(\theta) + (m_c + m_p) g \sin(\theta)}{\ell m_c + \ell m_p (1 - \cos^2(\theta))} \quad (1.2)$$

(a)

Denote $\mathbf{x} = [x, \theta, \dot{x}, \dot{\theta}, F]$ as the augmented state vector. The dynamics of cart-pole system can be written as follows:

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{x} \\ \dot{\theta} \\ \ddot{x} \\ \ddot{\theta} \\ \dot{F} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{\theta} \\ \frac{\ell m_p \sin(\theta) \dot{\theta}^2 + F + m_p g \cos(\theta) \sin(\theta)}{m_c + m_p (1 - \cos^2(\theta))} \\ -\frac{\ell m_p \cos(\theta) \sin(\theta) \dot{\theta}^2 + F \cos(\theta) + (m_c + m_p) g \sin(\theta)}{\ell m_c + \ell m_p (1 - \cos^2(\theta))} \\ \frac{u - F}{\tau_a} \end{bmatrix} = \mathbf{f}(\mathbf{x}, u) \quad (1.3)$$

- Forward Euler Discretization

Given the continuous-time nonlinear dynamics

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u), \quad (1.4)$$

the Forward Euler method with sampling time h approximates the state evolution as

$$\mathbf{x}_{k+1} = \mathbf{x}_k + h \mathbf{f}(\mathbf{x}_k, u_k). \quad (1.5)$$

For the augmented cart-pole state the discrete-time dynamics become

$$x_{k+1} = x_k + h \dot{x}_k, \quad (1.6)$$

$$\theta_{k+1} = \theta_k + h \dot{\theta}_k, \quad (1.7)$$

$$\dot{x}_{k+1} = \dot{x}_k + h \ddot{x}_k, \quad (1.8)$$

$$\dot{\theta}_{k+1} = \dot{\theta}_k + h \ddot{\theta}_k, \quad (1.9)$$

$$F_{k+1} = F_k + h \frac{u_k - F_k}{\tau_a}. \quad (1.10)$$

Therefore, the discrete nonlinear dynamics can be written compactly as

$$\mathbf{x}_{k+1} = F_d(\mathbf{x}_k, u_k) = \mathbf{x}_k + h \mathbf{f}(\mathbf{x}_k, u_k). \quad (1.11)$$

As an alternative, a higher-order numerical integration method such as the fourth-order Runge–Kutta (RK4) scheme may be used to improve integration accuracy.

(b)

Define $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]$ and $\mathbf{U} = [u_0, \dots, u_{N-1}]$ as matrices of state and control trajectory. Then a cost function for the swing-up task is as follows:

$$J(\mathbf{X}, \mathbf{U}) = \frac{1}{2}(\mathbf{x}_N - \mathbf{x}_{ref})^T \mathbf{Q}_N (\mathbf{x}_N - \mathbf{x}_{ref}) + \frac{1}{2} \sum_{k=0}^{N-1} ((\mathbf{x}_k - \mathbf{x}_{ref})^T \mathbf{Q} (\mathbf{x}_k - \mathbf{x}_{ref}) + Ru_k^2) \quad (1.12)$$

where

$$l_N(\mathbf{x}_N) = \frac{1}{2}(\mathbf{x}_N - \mathbf{x}_{ref})^T \mathbf{Q}_N (\mathbf{x}_N - \mathbf{x}_{ref}) \quad (1.13)$$

is the terminal cost and

$$\sum_{k=0}^{N-1} l_k(\mathbf{x}_k, u_k) = \frac{1}{2} \sum_{k=0}^{N-1} ((\mathbf{x}_k - \mathbf{x}_{ref})^T \mathbf{Q} (\mathbf{x}_k - \mathbf{x}_{ref}) + Ru_k^2) \quad (1.14)$$

is the running cost. Define the value function:

$$V(\mathbf{x}_k) = \min_{u_k, u_{k+1}, \dots, u_{N-1}} \left(\sum_{t=k}^{N-1} l_t(\mathbf{x}_t, u_t) + l_N(\mathbf{x}_N) \right) \quad (1.15)$$

The Bellman optimality condition provides a recursion on $V(\mathbf{x}_k)$ as follows:

$$V(\mathbf{x}_k) = \min_{u_k} (l_k(\mathbf{x}_k, u_k) + V(\mathbf{x}_{k+1})) \quad (1.16)$$

$$= \min_{u_k} (l_k(\mathbf{x}_k, u_k) + V_{k+1}(F_d(\mathbf{x}, u))) \quad (1.17)$$

(c)

Define the local action-value function $Q(\delta \mathbf{x}_k, \delta u_k)$ as follows:

$$Q(\delta \mathbf{x}_k, \delta u_k) = l_k(\mathbf{x}_k + \delta \mathbf{x}_k, u_k + \delta u_k) + V_{k+1}(F_d(\mathbf{x}_k + \delta \mathbf{x}_k, u_k + \delta u_k)) \quad (1.18)$$

Then we can approximate $Q(\delta \mathbf{x}_k, \delta u_k)$ in a quadratic form by expanding to 2-nd order of its Taylor expansion:

$$\begin{aligned} Q(\delta \mathbf{x}_k, \delta u_k) \approx Q(\mathbf{0}, 0) &+ Q_{x,k}^T \delta \mathbf{x} + Q_{u,k}^T \delta u_k + \frac{1}{2} (\delta \mathbf{x}_k^T Q_{xx,k} \delta \mathbf{x}_k + \delta u_k^T Q_{uu,k} \delta u_k) \\ &+ \delta u_k^T Q_{ux,k} \delta \mathbf{x}_k \end{aligned} \quad (1.19)$$

where from definition of 1.18 we can determine:

$$\begin{aligned} Q_k(0, 0) &= l(\mathbf{x}_k, u_k) + V_{k+1}(\mathbf{x}_{k+1}), \\ Q_x &= l_x + F_{dx}^T V_{x,k+1}, \\ Q_u &= l_u + F_{du}^T V_{x,k+1}, \\ Q_{xx} &= l_{xx} + F_{dx}^T V_{xx,k+1} F_{dx} + V_{x,k+1}^T F_{dxx}, \\ Q_{uu} &= l_{uu} + F_{du}^T V_{xx,k+1} F_{du} + V_{x,k+1}^T F_{duu}, \\ Q_{ux} &= l_{ux} + F_{du}^T V_{xx,k+1} F_{dx} + V_{x,k+1}^T F_{dux}. \end{aligned}$$

where the subscription $k+1$ indicate the value taken at time $k+1$. Otherwise we evaluate at time k .

(d)

Now we minimize $Q(\delta \mathbf{x}_k, \delta u_k)$ in 1.19 over δu_k , we have

$$\delta u_k = \arg \min_{\delta u_k} Q(\delta \mathbf{x}_k, \delta u_k) = \arg \min_{\delta u_k} \left[\frac{1}{2} \delta u_k^T Q_{uu,k} \delta u_k + Q_{u,k}^T \delta u_k + \delta u_k^T Q_{ux,k} \delta \mathbf{x}_k \right] \quad (1.20)$$

$$= -Q_{uu}^{-1} Q_{ux} \delta \mathbf{x} - Q_{uu}^{-1} Q_u^T \quad (1.21)$$

Defining the feedforward and feedback gains:

$$\mathbf{k}_k = -Q_{uu,k}^{-1} Q_{u,k}, \quad (1.22)$$

$$\mathbf{K}_k = -Q_{uu,k}^{-1} Q_{ux,k}, \quad (1.23)$$

the optimal perturbation is $\delta u_k^* = \mathbf{k}_k + \mathbf{K}_k \delta \mathbf{x}_k$.

Substituting $\delta u_k^* = \mathbf{k}_k + \mathbf{K}_k \delta \mathbf{x}_k$ back into the quadratic Q-function and collecting terms by order in $\delta \mathbf{x}_k$, using the identities $Q_{uu,k} \mathbf{k}_k = -Q_{u,k}$ and $Q_{uu,k} \mathbf{K}_k = -Q_{ux,k}$, gives the value-function update:

$$V_{x,k} = Q_{x,k} - \mathbf{K}_k^\top Q_{uu,k} \mathbf{k}_k, \quad (1.24)$$

$$V_{xx,k} = Q_{xx,k} - \mathbf{K}_k^\top Q_{uu,k} \mathbf{K}_k. \quad (1.25)$$

Terminal conditions. Since $V_N(\mathbf{x}_N) = l_N(\mathbf{x}_N) = \frac{1}{2}(\mathbf{x}_N - \mathbf{x}_{ref})^\top \mathbf{Q}_N(\mathbf{x}_N - \mathbf{x}_{ref})$, evaluating at the nominal terminal state $\bar{\mathbf{x}}_N$ gives:

$$V_{x,N} = \mathbf{Q}_N(\bar{\mathbf{x}}_N - \mathbf{x}_{ref}), \quad V_{xx,N} = \mathbf{Q}_N. \quad (1.26)$$

Backward pass algorithm.

1. **Initialize:** $V_{x,N} = \mathbf{Q}_N(\bar{\mathbf{x}}_N - \mathbf{x}_{ref})$, $V_{xx,N} = \mathbf{Q}_N$.

2. **For** $k = N - 1, N - 2, \dots, 0$:

(a) Compute Jacobians: $A_k = \partial F_d / \partial \mathbf{x} \big|_{\bar{\mathbf{x}}_k, \bar{u}_k}$, $B_k = \partial F_d / \partial u \big|_{\bar{\mathbf{x}}_k, \bar{u}_k}$.

(b) Evaluate Q-function derivatives:

$$\begin{aligned} Q_{x,k} &= l_{x,k} + A_k^\top V_{x,k+1}, & Q_{u,k} &= l_{u,k} + B_k^\top V_{x,k+1}, \\ Q_{xx,k} &= l_{xx,k} + A_k^\top V_{xx,k+1} A_k, & Q_{uu,k} &= l_{uu,k} + B_k^\top V_{xx,k+1} B_k, \\ Q_{ux,k} &= l_{ux,k} + B_k^\top V_{xx,k+1} A_k. \end{aligned}$$

(For full DDP, add $\sum_i V_{x,k+1}^{(i)} F_{dxx,k}^{(i)}$ to $Q_{xx,k}$, and analogous second-derivative contractions to $Q_{uu,k}$ and $Q_{ux,k}$.)

(c) Compute gains: $\mathbf{k}_k = -Q_{uu,k}^{-1} Q_{u,k}$, $\mathbf{K}_k = -Q_{uu,k}^{-1} Q_{ux,k}$.

(d) Update: $V_{x,k} = Q_{x,k} - \mathbf{K}_k^\top Q_{uu,k} \mathbf{k}_k$, $V_{xx,k} = Q_{xx,k} - \mathbf{K}_k^\top Q_{uu,k} \mathbf{K}_k$.

3. **Output:** Gain sequences $\{\mathbf{k}_k, \mathbf{K}_k\}_{k=0}^{N-1}$.

(e)

iLQR vs. DDP. iLQR discards the dynamics second-derivative tensors from the Q-function, retaining only first-order (Jacobian) contributions:

$$\begin{aligned} Q_{xx,k}^{\text{iLQR}} &= l_{xx} + A_k^\top V_{xx,k+1} A_k, \\ Q_{uu,k}^{\text{iLQR}} &= l_{uu} + B_k^\top V_{xx,k+1} B_k, \\ Q_{ux,k}^{\text{iLQR}} &= l_{ux} + B_k^\top V_{xx,k+1} A_k. \end{aligned}$$

Full DDP additionally includes the contraction $\sum_i V_{x,k+1}^{(i)} F_{dxx,k}^{(i)}$ in Q_{xx} , and analogous corrections in Q_{uu} and Q_{ux} , accounting for the curvature of the dynamics that iLQR ignores.

For the augmented cart-pole, the only nonzero dynamics second derivative with respect to the state is the curvature of the $\ddot{\theta}$ component in the θ direction:

$$\frac{\partial^2 f(\dot{\theta})}{\partial \theta^2} \propto \sin \theta, \quad (1.27)$$

arising from the terms $\cos \theta \sin \theta$ and $(m_c + m_p)g \sin \theta$ in the $\ddot{\theta}$ numerator. All other second partial derivatives of $\mathbf{f}$ with respect to state are zero.

The DDP-only correction to $Q_{xx,k}$ —a rank-1 update in the (θ, θ) slot—is:

$$\Delta Q_{xx,k}^{\text{DDP}} = \sum_i V_{x,k+1}^{(i)} F_{dxx,k}^{(i)} = h V_{x,k+1}^{(\dot{\theta})} \frac{\partial^2 f^{(\dot{\theta})}}{\partial \theta^2} \mathbf{e}_\theta \mathbf{e}_\theta^\top, \quad (1.28)$$

where $\mathbf{e}_\theta$ is the unit vector for the θ component of the augmented state.

This correction scales as $\sin \theta$ and vanishes at both equilibria ($\theta = 0$ and $\theta = \pi$, since $\sin \pi = 0$). It is largest at $\theta \approx \pi/2$ during swing-up. Retaining the DDP term is worthwhile when the trajectory sweeps through large pole angles, where the improved quadratic model accelerates convergence. Near the upright equilibrium the correction is negligible and iLQR provides essentially the same result at lower cost.

(f)

Forward pass and robustness. Given gains $\{\mathbf{k}_k, \mathbf{K}_k\}$ from the backward pass, the forward rollout with line-search step $\alpha \in (0, 1]$ is:

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0, \quad (1.29)$$

$$\delta u_k = \alpha \mathbf{k}_k + \mathbf{K}_k(\mathbf{x}_k - \bar{\mathbf{x}}_k), \quad (1.30)$$

$$u_k = \bar{u}_k + \delta u_k, \quad (1.31)$$

$$\mathbf{x}_{k+1} = F_d(\mathbf{x}_k, u_k), \quad k = 0, \dots, N-1. \quad (1.32)$$

The feedback term $\mathbf{K}_k(\mathbf{x}_k - \bar{\mathbf{x}}_k)$ corrects for deviations that accumulate when rolling out the nonlinear dynamics. The predicted cost decrease from the feedforward component is:

$$\Delta J(\alpha) = \alpha \sum_{k=0}^{N-1} Q_{u,k}^\top \mathbf{k}_k + \frac{\alpha^2}{2} \sum_{k=0}^{N-1} \mathbf{k}_k^\top Q_{uu,k} \mathbf{k}_k. \quad (1.33)$$

The step is accepted if $J_{\text{new}} \leq J_{\text{old}} + c_1 \Delta J(\alpha)$ for some $c_1 \in (0, 1)$ (Armijo condition). If rejected, α is reduced (e.g. $\alpha \leftarrow 0.5\alpha$) and the forward pass is repeated until acceptance or a minimum step size.

Levenberg–Marquardt regularization. When $Q_{uu,k} \neq 0$ —which can occur far from the solution or when DDP second-derivative terms are negative—the backward pass does not yield a descent direction. The regularization

$$Q_{uu,k} \leftarrow Q_{uu,k} + \mu \mathbf{I}, \quad \mu > 0, \quad (1.34)$$

with μ large enough to restore positive definiteness ensures that $\mathbf{k}_k = -(Q_{uu,k} + \mu \mathbf{I})^{-1} Q_{u,k}$ is a descent direction and limits the step magnitude, improving global robustness. In practice μ is adapted: increased when the actual-to-predicted cost ratio is low, decreased when steps are consistently accepted.

(g)

LQR special case. For linear dynamics $F_d(\mathbf{x}, u) = A\mathbf{x} + Bu$ (constant Jacobians, all second-derivative tensors zero) and quadratic cost $l(\mathbf{x}, u) = \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \frac{1}{2} R u^2$ with $\mathbf{x}_{\text{ref}} = \mathbf{0}$, we have $l_{xx} = \mathbf{Q}$, $l_{uu} = R$, $l_{ux} = \mathbf{0}$. Setting $P_k := V_{xx,k}$, the Q-function blocks become:

$$Q_{xx} = \mathbf{Q} + A^\top P_{k+1} A, \quad Q_{uu} = R + B^\top P_{k+1} B, \quad Q_{ux} = B^\top P_{k+1} A. \quad (1.35)$$

Substituting into the DDP value-function recursion gives:

$$P_k = \mathbf{Q} + A^\top P_{k+1} A - A^\top P_{k+1} B (R + B^\top P_{k+1} B)^{-1} B^\top P_{k+1} A, \quad (1.36)$$

with $P_N = \mathbf{Q}_N$. This is exactly the discrete-time finite-horizon LQR Riccati recursion.

Scalar backward step. Let $n = m = 1$ with $x_{k+1} = ax_k + bu_k$, $l = \frac{1}{2}qx^2 + \frac{1}{2}ru^2$, terminal cost $\frac{1}{2}p_N x_N^2$. One step backward from $k = N-1$:

$$\begin{aligned} Q_{xx} &= q + a^2 p_N, & Q_{uu} &= r + b^2 p_N, & Q_{ux} &= ab p_N, \\ K &= -\frac{ab p_N}{r + b^2 p_N}, \\ p_{N-1} &= Q_{xx} - K^2 Q_{uu} = q + a^2 p_N - \frac{a^2 b^2 p_N^2}{r + b^2 p_N} = q + \frac{a^2 r p_N}{r + b^2 p_N}, \end{aligned}$$

which is the standard scalar Riccati step.

(h)

Effect of actuator dynamics. The first-order lag $\dot{F} = (u - F)/\tau_a$ appends F as a state. Under Forward Euler the only nonzero control-coupling column is $B_k = (h/\tau_a) \mathbf{e}_F$, so the input contribution to Q_{uu} is:

$$Q_{uu} = R + \frac{h^2}{\tau_a^2} [V_{xx,k+1}]_{FF}. \quad (1.37)$$

Conditioning of Q_{uu} . For small τ_a the factor $h^2/\tau_a^2 \gg 1$ amplifies the value-function curvature in the force direction, making Q_{uu} large and potentially ill-conditioned relative to R ; Levenberg–Marquardt regularization becomes essential. For large τ_a , $h/\tau_a \rightarrow 0$ and $Q_{uu} \rightarrow R$, which is well-conditioned.

Achievable control bandwidth. The actuator introduces a first-order pole with time constant τ_a . The actual force $F(t)$ cannot track commands at frequencies above $\approx 1/\tau_a$ rad/s, bounding the achievable closed-loop bandwidth of the system.

Largest stable step h . Forward Euler applied to the actuator equation gives the discrete eigenvalue $\lambda = 1 - h/\tau_a$. Stability requires $|\lambda| < 1$, i.e., $h < 2\tau_a$. A smaller time constant therefore forces a finer integration step.

Actuator-free limit ($\tau_a \rightarrow 0$). As $\tau_a \rightarrow 0$, $F \rightarrow u$ instantaneously and the augmented 5-state model collapses to the standard 4-state cart-pole $[x, \theta, \dot{x}, \dot{\theta}]^\top$ with direct force input. While this simplifies the model, ignoring the physical lag means the controller commands inputs that the real actuator cannot track, degrading closed-loop performance when τ_a is non-negligible.

2 Problem 2: Constrained Cart-Pole: Model Predictive Control

Prerequisite: Linearization about the Upright Equilibrium

For brevity we drop the tilde and write $\mathbf{x}_k$ for the deviation state hereafter.

Let $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_{up}$ denote the deviation from the upright equilibrium $\mathbf{x}_{up} = [0, \pi, 0, 0]^\top$. Then from 1.3 we have:

$$\dot{\tilde{\mathbf{x}}} = \mathbf{f}(\mathbf{x}_{up} + \tilde{\mathbf{x}}, u) \quad (2.1)$$

Linearize the dynamic equation around $\mathbf{x}_{up}$ we have:

$$\dot{\tilde{\mathbf{x}}} \approx \mathbf{A}_c \tilde{\mathbf{x}} + \mathbf{B}_c u \quad (2.2)$$

where $\mathbf{A}_c = \partial \mathbf{f} / \partial \tilde{\mathbf{x}}|_{\mathbf{x}_{up}, 0}$ and $\mathbf{B}_c = \partial \mathbf{f} / \partial u|_{\mathbf{x}_{up}, 0}$.

Evaluation at the upright equilibrium. Let $\phi = \theta - \pi$ be the small angle deviation. Then $\sin \theta = \sin(\pi + \phi) \approx -\phi$ and $\cos \theta = \cos(\pi + \phi) \approx -1$ to first order. The denominator factor $D(\theta) = m_c + m_p \sin^2 \theta$ evaluates to $D(\pi) = m_c$. Setting $\phi, \dot{\phi}, u \rightarrow 0$ in the Jacobians of the $\ddot{x}$ and $\ddot{\theta}$ expressions gives:

$$\begin{aligned} \left. \frac{\partial f_{\ddot{x}}}{\partial \theta} \right|_{up} &= \left. \frac{m_p g (\cos^2 \theta - \sin^2 \theta)}{D} \right|_{\pi} = \frac{m_p g}{m_c}, & \left. \frac{\partial f_{\ddot{x}}}{\partial u} \right|_{up} &= \left. \frac{1}{D} \right|_{\pi} = \frac{1}{m_c}, \\ \left. \frac{\partial f_{\ddot{\theta}}}{\partial \theta} \right|_{up} &= \left. \frac{(m_c + m_p) g (-\cos \theta)}{\ell D} \right|_{\pi} = \frac{(m_c + m_p) g}{\ell m_c}, & \left. \frac{\partial f_{\ddot{\theta}}}{\partial u} \right|_{up} &= \left. \frac{-\cos \theta}{\ell D} \right|_{\pi} = \frac{1}{\ell m_c}. \end{aligned}$$

All other first partial derivatives of $\ddot{x}$ and $\ddot{\theta}$ vanish at the equilibrium (terms involve $\sin \pi = 0$, $\dot{\theta} = 0$, or $u = 0$). The remaining rows follow directly from $\dot{\tilde{x}}_1 = \tilde{x}_3$ and $\dot{\tilde{x}}_2 = \tilde{x}_4$. This yields:

$$\mathbf{A}_c = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{m_p g}{m_c} & 0 & 0 \\ 0 & \frac{(m_c + m_p) g}{\ell m_c} & 0 & 0 \end{bmatrix}, \quad \mathbf{B}_c = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{m_c} \\ \frac{1}{\ell m_c} \end{bmatrix}. \quad (2.3)$$

Note that both $\ddot{x}$ and $\ddot{\theta}$ are driven by the same two terms: the pole-tilt ϕ (through $m_p g$) and the input force F , both scaled by the total cart inertia m_c .

Discretization. Applying forward Euler with sample time T_s gives the discrete-time matrices:

$$A = I + T_s \mathbf{A}_c = \begin{bmatrix} 1 & 0 & T_s & 0 \\ 0 & 1 & 0 & T_s \\ 0 & \frac{m_p g T_s}{m_c} & 1 & 0 \\ 0 & \frac{(m_c + m_p) g T_s}{\ell m_c} & 0 & 1 \end{bmatrix}, \quad B = T_s \mathbf{B}_c = \begin{bmatrix} 0 \\ 0 \\ \frac{T_s}{m_c} \\ \frac{T_s}{\ell m_c} \end{bmatrix}, \quad (2.4)$$

(a)

The finite-horizon regulation MPC problem solved at each time step k , given the measured state $\hat{\mathbf{x}}_k$, is:

$$\begin{aligned} \min_{\{\mathbf{x}_i, u_i\}} \quad & \mathbf{x}_N^\top \mathbf{P} \mathbf{x}_N + \sum_{i=0}^{N-1} (\mathbf{x}_i^\top \mathbf{Q} \mathbf{x}_i + r u_i^2) \\ \text{subject to} \quad & \mathbf{x}_0 = \hat{\mathbf{x}}_k, \\ & \mathbf{x}_{i+1} = \mathbf{A} \mathbf{x}_i + \mathbf{B} u_i, \quad i = 0, \dots, N-1, \\ & |u_i| \leq u_{\max}, \quad i = 0, \dots, N-1, \\ & |[1 \ 0 \ 0 \ 0] \mathbf{x}_i| \leq x_{\max}, \quad i = 0, \dots, N-1, \\ & \mathbf{x}_N \in \mathcal{X}_f, \end{aligned} \tag{2.5}$$

where $\mathbf{P} \succeq 0$ is the terminal cost matrix, $\mathbf{Q} \succeq 0$ is the state weight, $r > 0$ is the input weight, and $\mathcal{X}_f$ is the terminal constraint set.

Receding-horizon algorithm.

1. At time step k : measure the current state $\hat{\mathbf{x}}_k$.
2. Solve problem (2.5) to obtain the optimal sequence $\{u_0^*, u_1^*, \dots, u_{N-1}^*\}$.
3. Apply only the first input: $u_k = u_0^*$.
4. Advance $k \leftarrow k + 1$, shift the horizon, and return to step 1.

(b)

Sparse QP form. Stack all states and inputs into a single decision vector:

$$z = [\mathbf{x}_0^\top, u_0, \mathbf{x}_1^\top, u_1, \dots, \mathbf{x}_{N-1}^\top, u_{N-1}, \mathbf{x}_N^\top]^\top \in \mathbb{R}^{(N+1)n + Nm}. \tag{2.6}$$

The cost is block-diagonal quadratic in z . The dynamics $\mathbf{x}_{i+1} = \mathbf{A} \mathbf{x}_i + \mathbf{B} u_i$ are enforced as Nn equality constraints, and the initial condition $\mathbf{x}_0 = \hat{\mathbf{x}}_k$ adds n more. The resulting QP has a block-banded (arrow-head) structure that is highly sparse.

Condensed QP form. Recursively substituting the dynamics eliminates all state variables. Every state can be written as:

$$\mathbf{x}_i = \mathbf{A}^i \hat{\mathbf{x}}_k + \sum_{j=0}^{i-1} \mathbf{A}^{i-1-j} \mathbf{B} u_j. \tag{2.7}$$

Stacking $\hat{\mathbf{X}} = [\mathbf{x}_0^\top, \mathbf{x}_1^\top, \dots, \mathbf{x}_N^\top]^\top$ and $\mathbf{U} = [u_0, u_1, \dots, u_{N-1}]^\top$:

$$\hat{\mathbf{X}} = \mathcal{A} \hat{\mathbf{x}}_k + \mathcal{B} \mathbf{U}, \tag{2.8}$$

where the prediction matrices are

$$\mathcal{A} = \begin{bmatrix} I \\ A \\ A^2 \\ \vdots \\ A^N \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} 0 & 0 & \dots & 0 \\ B & 0 & \dots & 0 \\ AB & B & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \dots & B \end{bmatrix}. \tag{2.9}$$

Defining $\bar{\mathbf{Q}} = \text{blkdiag}(\mathbf{Q}, \mathbf{Q}, \dots, \mathbf{Q}, \mathbf{P}) \in \mathbb{R}^{(N+1)n \times (N+1)n}$, the cost becomes a pure function of $\mathbf{U}$:

$$J = \mathbf{U}^\top \mathbf{H} \mathbf{U} + \mathbf{f}^\top \mathbf{U} + \text{const}, \tag{2.10}$$

with condensed Hessian and gradient:

$$\mathbf{H} = \mathcal{B}^\top \bar{\mathbf{Q}} \mathcal{B} + r \mathbf{I}_N \in \mathbb{R}^{N \times N}, \quad \mathbf{f} = \mathcal{B}^\top \bar{\mathbf{Q}} \mathcal{A} \hat{\mathbf{x}}_k \in \mathbb{R}^N. \tag{2.11}$$

The condensed QP is $\min_{\mathbf{U}} \frac{1}{2} \mathbf{U}^\top \mathbf{H} \mathbf{U} + \mathbf{f}^\top \mathbf{U}$ subject to $|u_i| \leq u_{\max}$ and position constraints expressed through $\mathcal{B}$.

(c)

For horizon N , n states, and $m = 1$ input:

Sparse form.

- Decision variables: $(N + 1)n + N = 5N + 4$ (for $n = 4$).
- Equality constraints: $(N + 1)n = 4N + 4$ (initial condition + N dynamics blocks).
- Inequality constraints: $2N$ (input bounds) + $2N$ (position bounds) = $4N$.

The constraint matrix is block-banded with $O(Nn)$ nonzero entries. Factorizing the banded KKT system costs $O(N)$, making this form scale linearly with horizon length.

Condensed form.

- Decision variables: N (inputs only; all states eliminated).
- Equality constraints: 0 (all dynamics absorbed into $\mathbf{H}, \mathbf{f}$).
- Inequality constraints: $2N$ (input) + $2N$ (position via $\mathcal{B}$) = $4N$.

The Hessian $\mathbf{H} \in \mathbb{R}^{N \times N}$ is dense (full because $\mathcal{B}^\top \bar{\mathcal{Q}} \mathcal{B}$ couples all time steps). Factorizing it costs $O(N^3)$.

As N grows the sparse form is superior: its banded KKT structure allows $O(N)$ solves, whereas the dense condensed Hessian incurs $O(N^3)$ cost. For small N (e.g., $N \leq 20$) the condensed form is often faster in practice due to lower overhead, but it becomes prohibitive for long horizons.

(d)

Although no closed-form expression $u = \kappa(\mathbf{x})$ is written, re-solving problem (2.5) at each measured state $\hat{\mathbf{x}}_k$ defines an implicit state-feedback law $u_k = u_0^*(\hat{\mathbf{x}}_k)$: the optimizer maps the current measurement to the optimal first input. The resulting closed-loop map is a piecewise-affine function of $\hat{\mathbf{x}}_k$ (from the parametric QP), even though it is never computed explicitly.

Relationship to LQR. As $N \rightarrow \infty$ with $\mathbf{P}$ set to the discrete-algebraic Riccati solution P_∞ (DARE solution for $(\mathbf{Q}, r, A, B)$), the MPC optimal value function converges to the infinite-horizon LQR value $V^{LQR}(\mathbf{x}) = \mathbf{x}^\top P_\infty \mathbf{x}$, and the first optimal input converges to the LQR gain:

$$u_0^* \xrightarrow{N \rightarrow \infty} -K_{LQR} \hat{\mathbf{x}}_k, \quad K_{LQR} = (r + B^\top P_\infty B)^{-1} B^\top P_\infty A. \quad (2.12)$$

For finite N , using $\mathbf{P} = P_\infty$ effectively extends the horizon to infinity, recovering the unconstrained LQR performance while the finite-horizon QP handles the active constraints.

(e)

Terminal conditions. For stability guarantees, the terminal cost $\mathbf{P}$ and terminal set $\mathcal{X}_f$ must satisfy:

1. There exists a local terminal controller $u = -K\mathbf{x}$ such that $A_{cl} = A - BK$ is stable.
2. $\mathbf{P}$ satisfies the Lyapunov equation: $A_{cl}^\top \mathbf{P} A_{cl} - \mathbf{P} = -(\mathbf{Q} + K^\top r K)$.
3. $\mathcal{X}_f \subseteq \mathcal{X}$ is positively invariant under A_{cl} : if $\mathbf{x} \in \mathcal{X}_f$ then $A_{cl}\mathbf{x} \in \mathcal{X}_f$, and $-K\mathbf{x}$ satisfies the input constraint.

Recursive feasibility (shift argument). Suppose at time k the optimal solution $\{u_0^*, u_1^*, \dots, u_{N-1}^*\}$ is feasible, with $\mathbf{x}_N^* \in \mathcal{X}_f$. At time $k + 1$, the candidate sequence

$$\{\tilde{u}_0, \tilde{u}_1, \dots, \tilde{u}_{N-1}\} = \{u_1^*, u_2^*, \dots, u_{N-1}^*, -K\mathbf{x}_N^*\} \quad (2.13)$$

is feasible for the shifted problem: the first $N - 1$ inputs are feasible by assumption; the appended terminal input $-K\mathbf{x}_N^*$ satisfies the input bound by condition 3; and the new terminal state $A_{cl}\mathbf{x}_N^* \in \mathcal{X}_f$ by the invariance condition. Hence the optimal $V^*(\mathbf{x}_{k+1}) \leq J(\text{candidate}) < \infty$, ensuring recursive feasibility.

Lyapunov decrease. Using the candidate sequence as an upper bound on $V^*(\mathbf{x}_{k+1})$:

$$\begin{aligned} V^*(\mathbf{x}_{k+1}) &\leq J(\mathbf{x}_{k+1}; \text{shifted}) \\ &= V^*(\mathbf{x}_k) - \ell(\mathbf{x}_k, u_0^*) + \underbrace{\ell(\mathbf{x}_N^*, -K\mathbf{x}_N^*) + V_f(A_{cl}\mathbf{x}_N^*) - V_f(\mathbf{x}_N^*)}_{\leq 0 \text{ by Lyapunov condition}}, \end{aligned} \quad (2.14)$$

where $V_f(\mathbf{x}) = \mathbf{x}^\top \mathbf{P} \mathbf{x}$ and the last bracket is ≤ 0 by condition 2. Therefore:

$$V^*(\mathbf{x}_{k+1}) - V^*(\mathbf{x}_k) \leq -\ell(\mathbf{x}_k, u_0^*) \leq 0, \quad (2.15)$$

establishing V^* as a Lyapunov function and the closed-loop system as asymptotically stable.

Terminal equality constraint. The simplest special case takes $\mathcal{X}_f = \{\mathbf{0}\}$ and $\mathbf{P} = 0$: the terminal constraint $\mathbf{x}_N = \mathbf{0}$ forces the trajectory to reach the origin within N steps, guaranteeing feasibility and stability (provided the system is controllable and N is large enough). The Lyapunov argument still holds since $V_f = 0$ drops all terminal cost terms.

(f)

NMPC formulation for swing-up. Let $\omega_i = \dot{\theta}_i$ and define the wrap-safe stage cost that is zero at the upright equilibrium ($\theta = \pi$, where $1 + \cos \pi = 0$) and equals 2 at the downward rest ($\theta = 0$):

$$\begin{aligned} \min_{\{\mathbf{x}_i, u_i\}} \quad & \sum_{i=0}^{N-1} [q_\theta(1 + \cos \theta_i) + q_\omega \omega_i^2 + r u_i^2] + V_f(\mathbf{x}_N) \\ \text{s.t.} \quad & \mathbf{x}_0 = \hat{\mathbf{x}}_k, \quad \mathbf{x}_{i+1} = F_d(\mathbf{x}_i, u_i), \quad |u_i| \leq u_{\max}, \quad |x_i| \leq x_{\max}. \end{aligned} \quad (2.16)$$

Direct multiple shooting introduces all states $\mathbf{x}_i$ and inputs u_i as decision variables ($N(n + m) + n$ total), and enforces the dynamics as defect equality constraints $\mathbf{x}_{i+1} - F_d(\mathbf{x}_i, u_i) = 0$. *Pro*: each shooting segment is integrated independently, enabling parallelism and easy warm-starting. *Con*: introduces Nn additional variables and equality constraints that increase NLP size.

Direct collocation places additional collocation points (e.g., Radau or Hermite-Simpson nodes) within each interval and enforces dynamics via polynomial interpolation constraints rather than explicit integration. *Pro*: can handle stiff dynamics and achieve high-order accuracy without small step sizes. *Con*: substantially more decision variables and constraint evaluations per interval; implementation is more complex.

Warm-starting by trajectory shifting. At time $k + 1$, initialize the NLP solver with the shifted previous solution: $\{\mathbf{x}_1^*, \dots, \mathbf{x}_N^*, F_d(\mathbf{x}_N^*, 0)\}$ and $\{u_1^*, \dots, u_{N-1}^*, 0\}$. This near-feasible guess typically reduces NLP iterations significantly compared to cold-starting.

Real-time iteration (RTI). Instead of solving the NLP to convergence, perform a single SQP/Newton step per sample: linearize the dynamics around the current guess, solve the resulting QP, and apply the first input immediately. The next step corrects the approximation error. This guarantees meeting the T_s deadline at the cost of slight suboptimality.

Connection to iLQR. The single Newton step in RTI amounts to solving a linear-quadratic subproblem. For the unconstrained (or input-bound-only) case, this linearization followed by a backward-forward DDP pass is precisely iLQR (Problem 1). With input saturation, a box-constrained variant of iLQR can be applied, making iLQR a natural inner solver for the real-time NMPC iteration.

(g)

Soft constraint formulation. Replace each hard track constraint $|[1\ 0\ 0\ 0]\mathbf{x}_i| \leq x_{\max}$ with a slack variable $\sigma_i \geq 0$ and add an ℓ_1 -exact penalty to the cost. The modified QP is:

$$\begin{aligned}
\min_{\{\mathbf{x}_i, u_i, \sigma_i\}} \quad & \mathbf{x}_N^\top \mathbf{P} \mathbf{x}_N + \sum_{i=0}^{N-1} (\mathbf{x}_i^\top \mathbf{Q} \mathbf{x}_i + r u_i^2 + \rho \sigma_i) \\
\text{s.t.} \quad & \mathbf{x}_0 = \hat{\mathbf{x}}_k, \quad \mathbf{x}_{i+1} = \mathbf{A} \mathbf{x}_i + \mathbf{B} u_i, \\
& |u_i| \leq u_{\max}, \\
& [1\ 0\ 0\ 0] \mathbf{x}_i \leq x_{\max} + \sigma_i, \\
& -[1\ 0\ 0\ 0] \mathbf{x}_i \leq x_{\max} + \sigma_i, \\
& \sigma_i \geq 0, \quad i = 0, \dots, N-1.
\end{aligned} \tag{2.17}$$

The problem remains a QP (cost is linear in σ_i). When ρ exceeds the optimal dual variable (Lagrange multiplier) of the original hard constraint, the ℓ_1 penalty is exact: the solution coincides with the hard-constrained solution whenever feasible, while permitting violation at a proportional cost when the hard constraint would render the problem infeasible.

Tube MPC. For systems subject to bounded additive disturbances $\mathbf{w}_k$ with $\|\mathbf{w}_k\| \leq \bar{w}$, tube MPC separates the problem into a nominal trajectory (optimized by MPC) and a local feedback controller $u = \kappa(\mathbf{x}) - K(\mathbf{x} - \mathbf{x}^{nom})$ that keeps the actual state within a precomputed invariant tube around the nominal. The nominal MPC is solved with tightened constraints (reduced by the tube cross-section), guaranteeing that the actual state satisfies the original constraints for all realizations of the disturbance, thereby providing robust recursive feasibility and stability.

(h)

QP size. For the condensed form with horizon N , $n = 4$ states, and $m = 1$ input:

- Decision variables: N (inputs $\mathbf{U}$) $+ N$ (slacks $\boldsymbol{\sigma}$, if soft) $= N$ to $2N$.
- Inequality constraints: $2N$ (input bounds) $+ 2N$ (position bounds) $= 4N$.
- Hessian: $N \times N$ dense, requiring $O(N^3)$ to factor (or $O(N^2)$ per active-set pivot).

For the sparse form, the QP has $5N + 4$ variables, $4N + 4$ equality constraints, and $4N$ inequalities, but the KKT matrix is block-banded with $O(N)$ nonzeros, allowing $O(N)$ factorization.

Solve-time vs. T_s requirement. The QP solution must be available within the sample interval T_s , so the solver wall-clock time must satisfy $t_{\text{solve}} < T_s$. For $N = 20$ – 50 and $n = 4$ the condensed QP involves a 20–50 variable dense QP solvable in microseconds to milliseconds with modern solvers (OSQP, quadprog), well within typical $T_s \approx 10$ – 50 ms for robotic systems. For NMPC with nonlinear dynamics, each Newton step additionally requires $O(N)$ sensitivity propagations.

Warm starting. Initializing the QP/NLP solver from the previous shifted solution means the active set rarely changes between steps. Active-set solvers then need only a few additional pivot operations (vs. many from cold-start), and interior-point methods converge in fewer iterations. Warm starting can reduce solve time by 5–10 $\times$, making real-time operation feasible at shorter T_s or longer horizons N .